

Michael Pignatelli

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EDUCATION

University of Pennsylvania, *School of Engineering and Applied Science*, Philadelphia, PA *Expected May 2027*

Bachelor of Science in Engineering in Computer Science and Cognitive Science (*Dual Major*)

Minor in Italian Culture | GPA: 3.51/4 | *Affiliations:* Wharton Data Science Group, ML Research at Penn, CS Society

Coursework: AI, TinyML, Data Structures & Algorithms, Operating Systems, Big Data Analytics, Probability (Wharton)

Master of Science in Engineering in Computer and Information Science (*Sub-matriculation*) *Expected May 2027*

EXPERIENCE

University of Pennsylvania | *Head Teaching Assistant – Computer and Information Science 1100* *August 2024 – Present*

- Lead and mentor **40+ undergraduate and graduate TAs** for Penn's largest intro CS course (~**300 students**), overseeing recitations, office hours, and grading; maintained TA quality rating of **3.25/4** – well above CIS dept. average.
- Develop and deliver instruction on data structures, recursion, OOP, functional programming, and Pandas.
- Guide students from “Hello, World!” through advanced topics to a final movie recommendation engine project.
- Maintain course infrastructure (cis1100.com) with HTML/YAML/CSS, reducing site errors and ensuring clarity.
- Collaborate with faculty to update curriculum, author exams, and lead weekly staff meetings.

Ava Labs (Avalanche) | *Software Engineering Intern – Platform Team* *June 2025 – August 2025*

- Implemented and productionized 4 gRPC APIs (**+7.5k LoC**) in the AvalancheGo blockchain client, replacing legacy JSON-RPC and enabling strongly-typed communication for all node operators. Reduced parsing errors and improved type safety, delivering **~50% lower latency** vs REST for **~1,800 node operators**.
- Wrote end-to-end and integration tests (Ginkgo) to validate gRPC API adoption and ensure cross-version stability.
- Authored incremental deployment plan adopted for H2 2025 rollout, ensuring backward compatibility. Aligned protocol changes with performance and security standards, contributing to long-term network stability across testnet/mainnet.
- Collaborated with Data Tooling and Cloudflare teams to align gRPC rollout with multi-team infrastructure standards.

Scale AI (via Outlier AI) | *Prompt Engineer (Contract)* *May 2024 – August 2024*

- Engineered and validated **500+ prompts** for LLM fine-tuning on math reasoning and algorithmic coding tasks, improving model accuracy and dataset consistency, across diverse math and programming benchmarks.
- Reviewed peer prompts and enforced quality standards, ensuring consistency across datasets and difficulty levels.

SELECTED PROJECTS

Tabby — Built full-stack bill-splitting app (**FastAPI, MongoDB, React**) using Google Gemini API; implemented tax/tip distribution, Venmo integration, payment tracking; demoed to **500+ peers** at **PennApps Hackathon** under 36-hour deadline.

Pedestrian Buddies — Built embedded CV system (**TensorFlow Lite, Keras**) for real-time crossing guard gesture detection; optimized inference for hardware-constrained systems, deployed on Arduino using TensorFlow Lite runtime [GitHub].

Housing & Education Analysis — Analyzed **1.6M+** U.S. housing & education records, including state-level educational spending; applied ridge regression, bootstrapping, and random forests to study spending–price relationships [Report].

Guess Who? — Developed Java Swing multiplayer game with client-server architecture and robust state sync [GitHub].

LEADERSHIP & TECHNICAL INVOLVEMENT

Penn High Powered Rocketry | *Active Recovery Engineering Lead/Head Safety Officer* *September 2023 – Present*

- Lead recovery system design for L3 rocket, implementing dual parachute deployment. Simulated flight dynamics (**<5% error vs actual**), contributing to successful **12,000 ft launch** at Spaceport America Cup/IREC (**120+ teams**).
- Conducted 3,600 ft field tests validating parachute energetics and deployment timing under live launch conditions.
- Integrated telemetry and comms testing with recovery hardware to ensure redundancy and reliable system performance.
- Oversee safety compliance for **30+ members** as Head Safety Officer, ensuring adherence to NAR/TRA regulations.

Penn Engineers without Borders | *Ramp Committee Engineer* *January 2025 – Present*

- Design ADA-compliant ramp in AutoCAD for Grace City Church, ensuring accessibility for **200+ weekly visitors**.

SKILLS

Programming & Frontend: Python, Go, Java, C/C++, OCaml, SQL, Swift, JS/TS, React, SwiftUI, HTML/CSS, Figma
Backend/Infra: gRPC/Protobuf, REST, FastAPI, MongoDB, Docker/Containerization, Git/GitHub (Actions), CI/CD, Cloudflare, Linux, Testing (JUnit, Ginkgo, pytest)

ML/Data Science: TensorFlow, Keras, PyTorch, Pandas, NumPy, RAG, Model Fine-Tuning, Prompt Engineering

APIs/Tools: Google Gemini, OpenAI, Anthropic, Jupyter, XML/JSON